

MACHINE LEARNING

LINEAR AND QUADRATIC DISCRIMINANT ANALYSIS

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Linear decision boundaries: hyperplanes

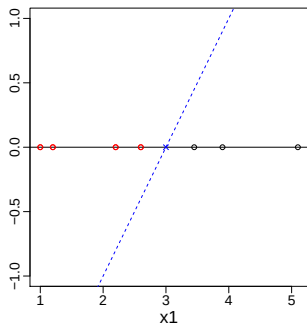
- ▶ Suppose we have only two classes, denoted by $\mathcal{G} = \{0, 1\}$.
- ▶ Linear classification methods produce **linear decision boundaries**, called **hyperplanes**.
- ▶ In a p -dimensional space, a hyperplane is a flat **affine subspace of dimension $p - 1$** , and it can be written, for parameters $b_0, b_1, \dots, b_p \in \mathbb{R}$, as the set

$$x \in \mathbb{R}^p : \quad b_0 + b_1 x_1 + \dots + b_p x_p = 0.$$

Example:

- ▶ For $p = 1$, the hyperplane is just a point

$$x \in \mathbb{R} : \quad b_0 + b_1 x = 0 \quad \Leftrightarrow \quad x = -b_0/b_1.$$



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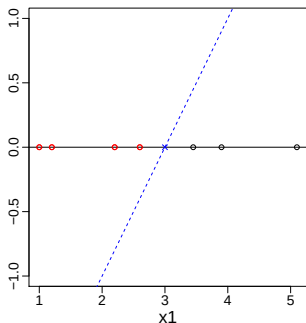
$$b_0 + b_1 x_1 + \dots + b_p x_p > 0,$$

tells us that it lies on the **one side** of the hyperplane, or with “ $<$ ” on the other.

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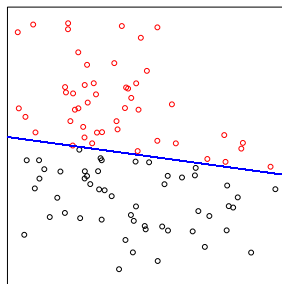
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Example:

- ▶ For $p = 2$, the hyperplane is a line

$$\begin{aligned}(x_1, x_2) \in \mathbb{R}^2 : \quad & b_0 + b_1 x_1 + b_2 x_2 = 0 \\ \Leftrightarrow \quad & x_2 = -b_0/b_2 - (b_1/b_2)x_1.\end{aligned}$$



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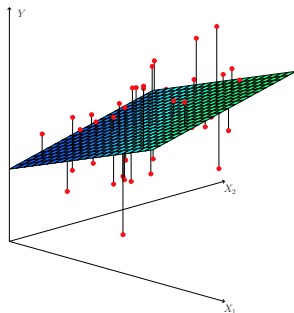
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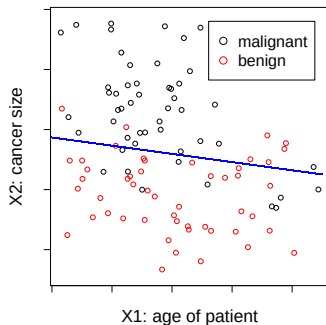
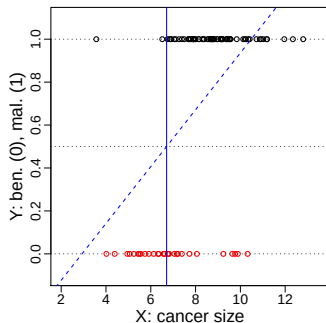
- ▶ For $p = 3$, the hyperplane is

$$(x_1, x_2, x_3) \in \mathbb{R}^3 : \quad b_0 + b_1 x_1 + b_2 x_2 + b_3 x_3 = 0.$$



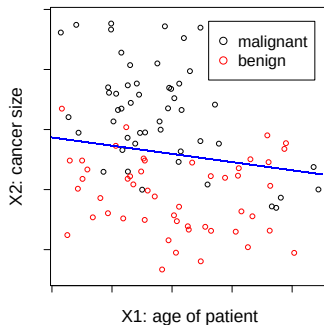
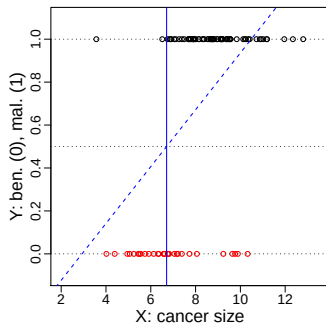
Linear methods for classification

- Example: direct linear regression on $\mathcal{G} = \{0, 1\}$; usually not a good idea!



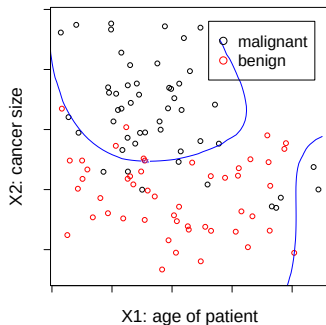
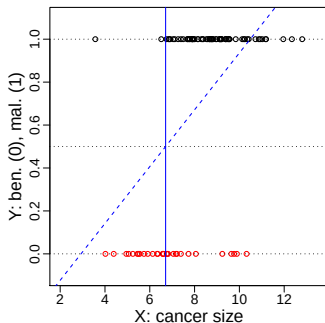
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- ▶ Many other methods will produce linear decision boundaries.



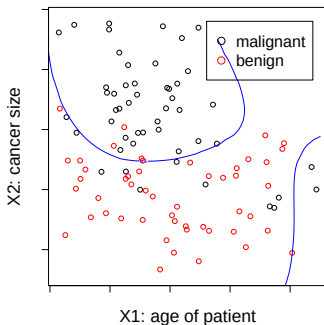
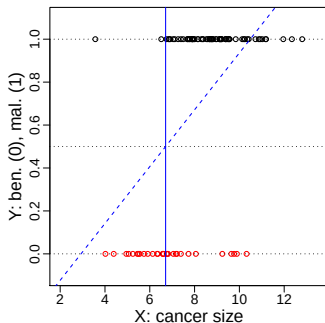
Linear methods for classification

- ▶ Example: direct linear regression on $\mathcal{G} = \{0, 1\}$; usually not a good idea!
- ▶ Many other methods will produce linear decision boundaries.
- ▶ Decision boundaries of linear methods are linear only in the **feature space**.
- ▶ **Augmenting the feature** space by basis functions might lead to non-linear decision boundaries in the original space of $(X_1, \dots, X_p)$.
- ▶ Right: the model is linear in $X_1, X_2, X_1^2, X_2^2, X_1^3, X_2^3, X_1X_2, X_1X_2^2, X_1^2X_2, X_1^2X_2^2$.



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- ▶ For more than two classes, linear classification methods produce **piecewise linear decision boundaries**.



Linear discriminant analysis

- ▶ **Setup:** Classify $Y \in \mathcal{G} = \{1, \dots, q\}$ given predictors $X = (X_1, \dots, X_p)$
- ▶ **Idea:** model the **class-conditional densities** g_j of $X \mid Y = j$, and the marginals $\pi_j = \Pr(Y = j)$, $j = 1, \dots, q$. Deduce $\Pr(Y \mid X)$ from **Bayes' formula**:

$$\Pr(Y = j \mid X = x) = \frac{\Pr(X = x \mid Y = j)\Pr(Y = j)}{\Pr(X = x)} \propto g_j(x)\pi_j.$$

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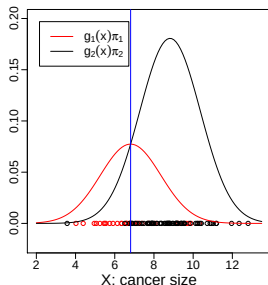
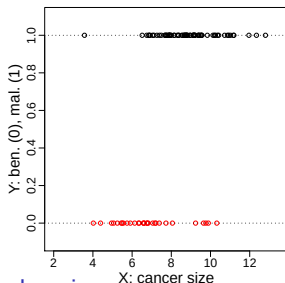
$$\Pr(Y = j \mid X = x) = \frac{\Pr(X = x \mid Y = j)\Pr(Y = j)}{\Pr(X = x)} \propto g_j(x)\pi_j.$$

- Linear discriminant analysis (LDA):

- (1) model $g_j(x)$ by a **multivariate normal distribution**,

$$g_j(x) = \frac{1}{(2\pi)^{p/2}(\det \Sigma_j)^{1/2}} \exp \left\{ -\frac{1}{2}(x - \mu_j)^T \Sigma_j^{-1}(x - \mu_j) \right\};$$

- (2) assume the g_j have the **same covariance matrix** $\Sigma_j = \Sigma$ for all j .



► **Model parameters:**

- the class/prior probabilities $\pi_1, \dots, \pi_q \in (0, 1)$;
- the mean vectors $\mu_1, \dots, \mu_q \in \mathbb{R}^P$;
- the covariance matrix $\Sigma \in \mathbb{R}^{P \times P}$.

- **Estimation:** From a training sample $(y_1, x_1), \dots, (y_n, x_n)$ we estimate the parameters of LDA by the usual **maximum likelihood estimators**

$$\begin{aligned}\hat{\pi}_j &= \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{y_i = j\} \\ \hat{\mu}_j &= \frac{\sum_{i=1}^n x_i \mathbf{1}\{y_i = j\}}{\sum_{i=1}^n \mathbf{1}\{y_i = j\}} \\ \hat{\Sigma} &= \frac{\sum_{j=1}^q \sum_{i=1}^n (x_i - \hat{\mu}_j)(x_i - \hat{\mu}_j)^T \mathbf{1}\{y_i = j\}}{n - q}\end{aligned}$$

- This is very simple and **computationally fast**!
- Note: Different to linear regression, we don't use the numerical value of the y_i 's but only the class!

LDA: prediction

- ▶ **Goal:** for a new input value $x_0 \in \mathbb{R}^p$ predict its unobserved class y_0 .
- ▶ **Method:** compute the **posterior class probabilities** $\widehat{\Pr}(Y = j \mid X = x_0)$ and use the **Bayes classifier** resulting from these estimates (plug-in estimate), i.e.,

$$\hat{y}_0 = \arg \max_{j=1, \dots, q} \widehat{\Pr}(Y = j \mid \mathbf{X} = x_0) = \arg \max_{j=1, \dots, q} \log \widehat{\Pr}(Y = j \mid X = x_0).$$

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- Note that

$$\begin{aligned} \log \widehat{\Pr}(Y = j \mid X = x_0) &= \log \widehat{g}_j(x_0) + \log \widehat{\pi}_j + \text{constant} \\ &= -\frac{1}{2}(x_0 - \widehat{\mu}_j)^T \widehat{\Sigma}^{-1}(x_0 - \widehat{\mu}_j) + \log \widehat{\pi}_j + \text{constant} \\ &= x_0^T \widehat{\Sigma}^{-1} \widehat{\mu}_j - \frac{1}{2} \widehat{\mu}_j^T \widehat{\Sigma}^{-1} \widehat{\mu}_j + \log \widehat{\pi}_j + \text{constant}, \end{aligned} \tag{1}$$

where the constant includes terms that do not depend on j .

- **Geometric interpretation:** eq. (1) implies that LDA classifies x_0 in terms of the **Mahalanobis distance**, $d_{\widehat{\Sigma}^{-1}}(x, y) = x^T \widehat{\Sigma}^{-1} y$, of x_0 from the centers $\widehat{\mu}_1, \dots, \widehat{\mu}_q$ (with a correction for the prior class probabilities).

LDA: decision boundaries

- The **decision boundary** that separates two classes $j, m \in \{1, \dots, q\}$ are all $x \in \mathbb{R}^p$ with

$$\Pr(Y = j \mid X = x) = \Pr(Y = m \mid X = x),$$

which is equivalent to

$$\log \Pr(Y = j \mid X = x) - \log \Pr(Y = m \mid X = x) = 0.$$

Replacing the probabilities by their expressions in terms of the Gaussian densities and the prior class probabilities we get the decision boundary

$$\log \frac{\pi_j}{\pi_m} - \frac{1}{2}(\mu_j + \mu_m)^T \Sigma^{-1}(\mu_j - \mu_m) + x^T \Sigma^{-1}(\mu_j - \mu_m) = 0,$$

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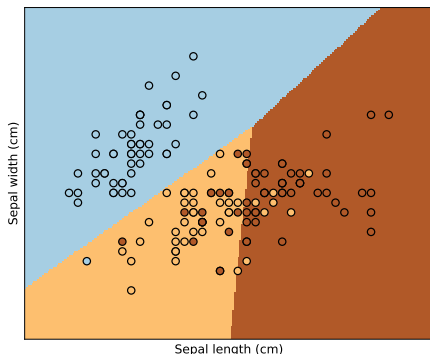
- ▶ Hence the decision boundaries separating any two classes j and m are hyperplanes in $\mathbb{R}^p$ of dimension $p - 1$.
- ▶ Thus the decision boundary that separates the q classes is **piecewise linear**.

Iris data set

For 150 measured plants; $p = 4$ predictors as response the three Iris species

$\mathcal{G} = \{\text{setosa}, \text{versicolor}, \text{virginica}\}$ the plant belongs. We only use two predictors here.

```
from sklearn.discriminant_analysis import LinearDiscriminantAnalysis
iris = sklearn.datasets.load_iris()
X, y = iris.data[:, :2], iris.target # we only keep the first two features: 'Sepal length', 'Sepal width'
lda = LinearDiscriminantAnalysis()
lda.fit(X, y)
x_min, x_max = X[:, 0].min() - .5, X[:, 0].max() + .5
y_min, y_max = X[:, 1].min() - .5, X[:, 1].max() + .5
h = .02 # step size in the mesh
xx, yy = np.meshgrid(np.arange(x_min, x_max, h), np.arange(y_min, y_max, h))
Z = lda.predict(np.c_[xx.ravel(), yy.ravel()]).reshape(xx.shape)
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Extensions of LDA

- There are many possibilities to extend LDA. The Bayes' construction

$$\Pr(Y = j \mid X = x) = \frac{\Pr(X = x \mid Y = j)\Pr(Y = j)}{\Pr(X = x)} = \frac{g_j(x)\pi_j}{\sum_{j=1}^q g_j(x)\pi_j}.$$

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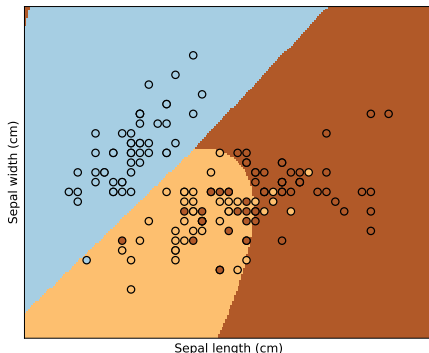
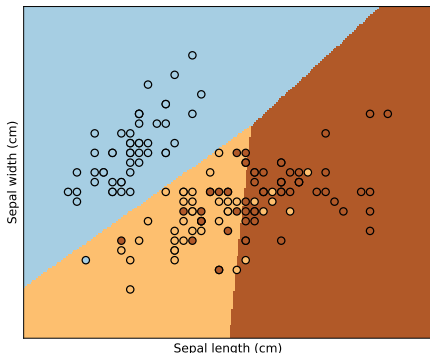
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- ▶ **Quadratic discriminant analysis** (QDA): model g_j by multivariate normals with different covariance matrices Σ_j .
- ▶ **Mixture** models: use mixtures of Gaussians for the g_j .
- ▶ **Naive Bayes**: conditional independence assumption, $g_j(x) = \prod_{l=1}^p g_{j,l}(x_l)$.
- ▶ When some features are **categorical**, for example X_1 , we can use

$$g_j(x) = \Pr(X_1 = x_1 \mid Y = j)g_j(x_2, \dots, x_p \mid x_1),$$

where $g_j(x_2, \dots, x_p \mid x_1)$ is a density for $X_2, \dots, X_p \mid X_1 = x_1, Y = j$.

- ▶ **Non-parametric** approaches: estimate g_j non-parametrically.

Comments on LDA

- ▶ LDA is easy and fast to compute.
- ▶ LDA provides estimates of the posterior class probabilities

$$\widehat{\Pr}(Y = j \mid X = x_0) = \frac{\widehat{g}_j(x_0)\widehat{\pi}_j}{\sum_{j=1}^q \widehat{g}_j(x_0)\pi_j}.$$

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- ▶ LDA usually performs well and can compete with more complex methods.
- ▶ Why? Clearly not because the normality assumption is realistic, but because a linear decision boundary is often the best the data can support and LDA is a **stable estimation method**.
- ▶ We don't need to have good models for the g_j to obtain a good model for the posterior: complex g_j may not be visible in the decision (see figure).

